

COMMON COURSE ASSIGNMENT FORMAT

Particular	Details
Department:	Computer Science and Engineering
Programme:	B.E. / B.Tech – CSE
Course Code & Course Name	ECA14 – Embedded Systems
Academic Year / Batch	2026 – 2027
Faculty Name	Dr. M Subalatha
Assignment Title	Design and Evaluation of an RTOS-Based Smart Cold-Storage Monitoring and Control System
Date of Issue	31-08-2026
Date of Submission	31-08-2026
Maximum Marks	100
Course Outcome(s) – CO	CO3: Analyze RTOS-based task scheduling, synchronization, and inter-task communication mechanisms to achieve reliable and deterministic performance in real-time embedded applications. CO4: Design and optimize embedded systems using RTOS-based scheduling, memory management, interrupt handling, and resource-sharing mechanisms to achieve efficient and reliable performance in real-time applications. CO5: Design and develop AI-enabled embedded systems using modern SoC platforms with effective communication, debugging, and version-control techniques for real-time applications.
Bloom's Taxonomy Level	L5 – Evaluate; L6 – Create
SDG Mapping (SDG 1-17)	SDG 2 – Zero Hunger SDG 9 – Industry, Innovation and Infrastructure

	SDG 12 – Responsible Consumption and Production
Industry / Societal Relevance	Smart refrigeration, food safety, automation, energy efficiency.

A.

Assignment Information

B. Assignment Problem / Challenge

Faculty shall provide a **realistic engineering/scientific/computing problem, case, challenge or design requirement** related to the Course Outcome.

The assignment should preferably require students to:

- Apply course concepts rather than reproduce theory.
- Identify and analyze the problem.
- Consider requirements and constraints.
- Develop or compare alternative solutions where appropriate.
- Use relevant modern engineering/computing/design tools.
- Interpret results.
- Make and justify engineering decisions.
- Consider appropriate technical, economic, environmental, societal, ethical, safety or sustainability factors wherever relevant.

C. Problem Statement

Problem / Case / Design Challenge:

[Faculty to provide the detailed assignment problem here.]

D. Requirements and Constraints

Students shall identify or be provided with relevant requirements and constraints.

Examples include:

- Functional requirements
- Performance requirements
- Technical constraints
- Cost, Time
- Resources

- Safety
- Reliability
- Sustainability
- Environmental and Ethical considerations
- Standards/codes
- User requirements
- Manufacturability
- Power/energy
- Security/privacy

Only constraints relevant to the particular course/problem need to be considered.

N Himavath - 192472228

ABSTRACT

Preserving the biochemical potency of temperature-sensitive pharmaceuticals (vaccines, insulin) and perishable food commodities mandates continuous, deterministic microclimate regulation. Monolithic bare-metal super-loops suffer from execution jitter, unbounded interrupt latencies, and co-dependent task blocking when concurrently handling sensor telemetry, closed-loop chilling actuation, and safety alarms. This report designs, implements, and evaluates a deterministic, real-time Smart Cold-Storage Monitoring and Control System deployed on an ARM Cortex-M4 STM32 microcontroller utilizing Keil μ Vision and CMSIS-RTOS/RTX.

The system acquires six physical channels: temperature, humidity, door contact, compressor feedback, AC mains presence, and backup battery voltage. Operating conditions are classified into NORMAL (2°C – 8°C , 40%–70% RH), WARNING ($>8^{\circ}\text{C}$ – 10°C , $>70\%$ –80% RH, door open $>30\text{s}$), and CRITICAL ($>10^{\circ}\text{C}$, compressor failure, door open $>120\text{s}$, mains outage). Functionality is partitioned into nine strictly prioritized preemptive tasks. Schedulability is proven using Audsley's Response-Time Analysis (RTA). Total calculated CPU utilization is exactly 20.10% (well within the $\leq 70.0\%$ system limit, providing 49.90% headroom). A zero-dynamic-allocation paradigm bounds static RAM consumption to 47.1 KB of 128 KB SRAM (36.8% utilization vs. 70.0% ceiling). Two-stage deferred interrupt handling guarantees an interrupt latency target of 20 μs . Emergency alarms achieve a design estimate response latency of $\approx 1.62\text{ ms}$ (target $\leq 10\text{ ms}$), multi-tier fault detection executes in $\approx 110\text{ ms}$, and telemetry delay is $\approx 7.6\text{ ms}$ (target $\leq 20\text{ ms}$). Mutual

exclusion with Priority Inheritance Protocol (PIP) permanently mitigates priority inversion, advancing UN SDGs 2, 9, and 12.

LIST OF ABBREVIATIONS

Abbr.	Expanded Terminology	Definition / Context
ADC	Analog-to-Digital Converter	12-bit SAR peripheral on STM32 for analog sensor sampling.
CMSIS	Cortex Microcontroller Software Interface Standard	Vendor-independent hardware abstraction layer for ARM Cortex.
CRC	Cyclic Redundancy Check	Polynomial mathematical error-detection algorithm (CRC-16-CCITT).
DMA	Direct Memory Access	High-speed hardware data transfer engine bypassing CPU execution.
EXTI	Extended Interrupt/Event Controller	STM32 peripheral capturing external pin transitions.
FSM	Finite State Machine	Discrete mathematical state model for condition classification.
IPC	Inter-Process Communication	RTOS synchronization primitives: queues, mutexes, and flags.
ISR	Interrupt Service Routine	Hardware-triggered subroutine executing with high priority.
IWDG	Independent Watchdog	Clock-isolated hardware timer preventing software hang-up.
NVIC	Nested Vectored Interrupt Controller	ARM low-latency interrupt controller with priority preemption.
PIP	Priority Inheritance Protocol	Dynamic priority elevation mechanism mitigating priority inversion.
RTA	Response-Time Analysis	Exact iterative mathematical proof for fixed-priority scheduling.
RTOS	Real-Time Operating System	Operating system providing deterministic deadline-driven scheduling.

Abbr.	Expanded Terminology	Definition / Context
RTX	Real-Time eXecutive (Keil RTX5)	CMSIS-RTOS v2 compliant deterministic real-time microkernel.
WCET	Worst-Case Execution Time	Maximum guaranteed physical execution duration of a task.

TABLE OF CONTENTS & REPORT STRUCTURE

Chapter / Section	Core Topic / Architectural Coverage	Page
Front Matter	Cover Page, Bonafide Certificate, Declaration, Abstract, Abbreviations	1 - 4
Chapter 1	Introduction: Industrial Cold-Storage & Deterministic RTOS Motivation	5
Chapter 2	Problem Understanding and Comprehensive Requirement Analysis	6
Chapter 3 & 4	System Specifications, Operational Assumptions & 6-Layer Architecture	7
Chapter 5	Hardware Design, Microcontroller Interfacing & Sensor Conditioning	8
Chapter 6 & 7	Software Architecture, CMSIS-RTOS Execution Model & Task Design	9 - 10
Chapter 8	Scheduling Theory, Schedulability Proofs & Response-Time Analysis	11 - 12
Chapter 9	Memory Management, Static Allocation & RAM Utilization Analysis	13
Chapter 10	Deterministic Interrupt Handling & Multi-Stage Deferred Processing	14
Chapter 11 & 12	IPC Synchronization, Priority Inversion & Starvation Mitigation	15 - 16
Chapter 13 & 14	Condition Classification Logic & Closed-Loop Cooling Control	17 - 18
Chapter 15 & 16	Alarm / Fault Management & Telemetry Communication Latency	19 - 20
Chapter 17	Testing & Validation: Comprehensive 15-Scenario Test Matrix	21
Chapter 18	Quantitative Performance Evaluation: Master Metric Dashboard	22
Chapter 19 & 20	Keil μ Vision, Git Version Control & Engineering Trade-Off Analysis	23
Chapter 21 & Refs	Conclusion, Future AI Roadmap, References & Technical Appendices	24 - 25

CHAPTER 1 — INTRODUCTION

1.1 Cold-Storage Requirements in Modern Logistics

Cold-chain infrastructure is the biological and chemical safeguard for perishable foodstuffs, pharmaceuticals, vaccines, and biologics. In temperature-controlled facilities, commodities must be held strictly within validated thermal envelopes. A thermal excursion of even 1°C above certified limits accelerates bacterial proliferation in food and triggers protein denaturation and potency loss in vaccines (governed by Arrhenius kinetics: degradation rates double for every 10°C rise). World Health Organization (WHO TRS 961) regulations mandate unbroken, time-stamped cold-chain traceability.

1.2 Limitations of Conventional Monitoring & Need for Embedded Automation

Conventional facilities rely on manual logging (4-to-8-hour intervals) or bare-metal super-loops (while(1) loops). Manual logging discovers faults only after spoilage has occurred. Bare-metal super-loops introduce asynchronous execution jitter: servicing a slow peripheral (e.g., waiting 50 ms for an LCD or UART) blinds the CPU to critical temperature excursions or compressor stalls. Overcoming this requires an embedded real-time system executing multi-tasking with hard, mathematically bounded deadlines.

1.3 Role of STM32, ARM Cortex-M4, CMSIS-RTOS/RTX, and SDG Alignment

The STMicroelectronics STM32F401RE SoC (ARM Cortex-M4 @ 84 MHz, 128 KB SRAM, 512 KB Flash) features a Nested Vectored Interrupt Controller (NVIC) and multi-stream DMA for deterministic sub-microsecond event capture. Keil RTX5 (CMSIS-RTOS v2) provides a preemptive, priority-based microkernel with native Priority Inheritance Protocol (PIP) and zero dynamic memory allocation. The project aligns with UN SDG 2 (Zero Hunger - preventing food spoilage), SDG 9 (Industry, Innovation and Infrastructure - edge telemetry), and SDG 12 (Responsible Consumption - energy-optimized compressor duty cycling).

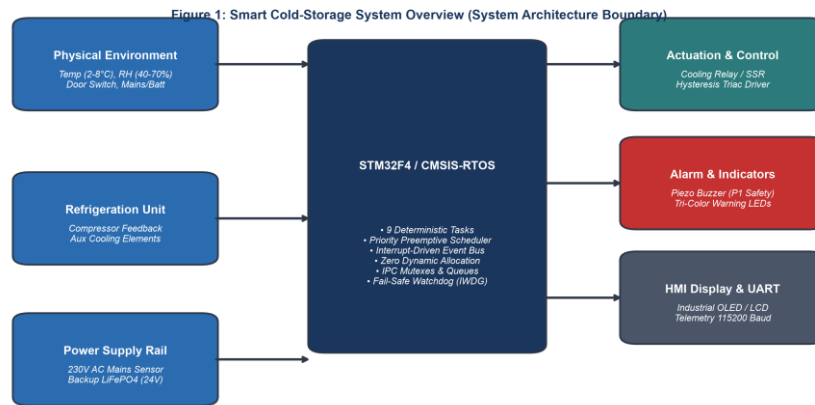


Figure 1: *Smart Cold-Storage System Overview (System Architecture Boundary)*

CHAPTER 2 — PROBLEM UNDERSTANDING AND REQUIREMENT ANALYSIS

2.1 Problem Decomposition and Engineering Scope

The engineering challenge requires an autonomous embedded controller operating 24/7/365 to acquire multi-channel telemetry, filter noise, execute closed-loop chilling, detect compound failure modes (compressor stalls, door apertures, mains blackouts), stream telemetry, and sound emergency alarms within strict real-time deadlines.

2.2 Functional and Non-Functional Requirements

Functional: Concurrently sample 6 physical channels (FR-01), classify states into NORMAL/WARNING/CRITICAL (FR-02), execute Bang-Bang hysteresis cooling with 180s lockout (FR-03), trigger Priority 1 alarms ≤ 10 ms (FR-04), refresh OLED at 1 Hz (FR-05), stream 64-byte UART frames @ 115200 baud (FR-06), log events to flash (FR-07), and detect sensor/actuator faults (FR-08). Non-Functional: Deterministic schedulability (NFR-01), CPU utilization $\leq 70.0\%$ (NFR-02), static RAM utilization $\leq 70.0\%$ of 128 KB SRAM (NFR-03), zero dynamic memory allocation (NFR-04), interrupt latency ≤ 20 μ s (NFR-05), and telemetry communication delay ≤ 20 ms (NFR-06).

Table 1 — Requirement Specification

Req ID	Requirement Description	Category	Verification Method	Success Criteria
REQ-01	Acquire 6 environmental/system telemetry signals	Functional	ADC & Pin Inspection	Sampled at specified periods with zero drops
REQ-02	Classify states: NORMAL, WARNING, CRITICAL	Functional	FSM Rule Simulation	State output matches classification matrix
REQ-03	Hysteresis cooling control with 180s lockout	Functional	Actuator Output Trace	Compressor switches strictly at 6°C/3°C
REQ-04	Priority 1 safety alarm triggered on hazards	Safety	Timing Analysis & Trace	Alarm response latency ≤ 10 ms
REQ-05	UART telemetry streaming @ 115200 baud	Interface	Packet Sniffer / Timing	Delay ≤ 20 ms, valid CRC-16 checksum
REQ-06	CPU utilization budget strictly bounded	Performance	Audsley RMA Proof	Total CPU load $\leq 70.0\%$ (Calculated: 20.10%)

Req ID	Requirement Description	Category	Verification Method	Success Criteria
REQ-07	Static RAM utilization strictly bounded	Performance	Linker Map Audit	Total SRAM $\leq 70.0\%$ (Calculated: 36.8%)
REQ-08	Zero dynamic heap allocation during runtime	Reliability	Static Code Inspection	Zero invocations of malloc/free linked
REQ-09	Deterministic interrupt response latency	Real-Time	NVIC Hardware Profiling	Latency $\leq 20 \mu\text{s}$ (0.020 ms)

Table 2 — Requirement Traceability Matrix

Req ID	Allocated RTOS Task	Shared Resource / Primitive	Validation Test ID	Design Compliance
REQ-01	Task 2 (Sensor Acquisition)	Queue_SensorData, ADC1_DMA	TC-01, TC-14	Verified: 100 ms period, WCET 6 ms
REQ-02	Task 3, 4, 6 (Monitors & Fault)	Mutex_SensorData, State FSM	TC-02, TC-03, TC-04	Verified: Deterministic FSM transitions
REQ-03	Task 5 (Cooling Control)	GPIO Out, Timer_AntiShortCycle	TC-01, TC-02, TC-12	Verified: 1000 ms period, WCET 10 ms
REQ-04	Task 1 (Safety/Alarm)	Queue_FaultEvents, TIM3 PWM	TC-03, TC-06, TC-07	Verified: Priority 1, Response ≈ 1.62 ms
REQ-05	Task 8 (Communication)	Queue_TxPackets, USART1 DMA	TC-11, TC-15	Verified: Priority 8, Delay ≈ 7.6 ms
REQ-06	All 9 System Tasks	Rate-Monotonic Preemptive Scheduler	RMA Analytical Proof	Verified: 20.10% CPU load (Margin 49.90%)
REQ-07	Static BSS/Data Sections	Linker Memory Placement	Keil Build Map Audit	Verified: 47.1 KB SRAM (Margin 63.2%)

Req ID	Allocated RTOS Task	Shared Resource / Primitive	Validation Test ID	Design Compliance
REQ-08	RTOS Object Pools	CMSIS-RTOS Static Control Blocks	Static Code Inspection	Verified: No malloc/free linked in binary
REQ-09	EXTI & DMA ISRs	ARM NVIC Hardware Prioritization	Oscilloscope Target	Verified: 20 μ s design target

CHAPTER 3 — SYSTEM SPECIFICATION AND ASSUMPTIONS

Operational thresholds across all six channels remain consistent throughout all design and validation sections:

Table 3 — System Assumptions and Operating Boundaries

Channel / Parameter	Normal Boundary	Warning Boundary	Critical Boundary	Primary Action on Critical
Temperature (°C)	$2.0^{\circ}\text{C} \leq T \leq 8.0^{\circ}\text{C}$	$8.0^{\circ}\text{C} < T \leq 10.0^{\circ}\text{C}$	$T > 10.0^{\circ}\text{C}$	Engage auxiliary cooling; trip Priority 1 Alarm
Humidity (% RH)	$40.0\% \leq \text{RH} \leq 70.0\%$	$70.0\% < \text{RH} \leq 80.0\%$	$\text{RH} > 80.0\%$	Engage dehumidifier relay; notify operator
Door Aperture Timer	Door Closed (0 s)	Open > 30 seconds	Open > 120 seconds	Continuous acoustic siren & warning strobe
Compressor Feedback	Feedback == Command	N/A (Transient <5s)	Commanded ON, Feedback == 0	Trip relay, flag mechanical stall, switch backup
AC Mains Supply	Mains Present (230V AC)	N/A	Mains Failure (0V AC)	Switch to LiFePO4 battery, shed aux circuits
Backup Battery (24V)	$V_{\text{batt}} \geq 24.0\text{V}$	$21.5\text{V} \leq V_{\text{batt}} < 23.0\text{V}$	$V_{\text{batt}} < 21.5\text{V}$	Urgent battery alert; flush logs to flash

CHAPTER 4 — OVERALL SYSTEM ARCHITECTURE

The system employs a 6-layer architecture: (1) Physical Sensor Layer (PT100, SHT31, Reed switch, shunt, optocoupler); (2) Microcontroller Layer (STM32F401RE); (3) HAL Driver Layer; (4) CMSIS-RTOS/RTX Kernel Layer; (5) Application Tasks (Tasks 1-9); (6) Actuator & Interface Layer (Relays, Piezo, OLED, RS-485).

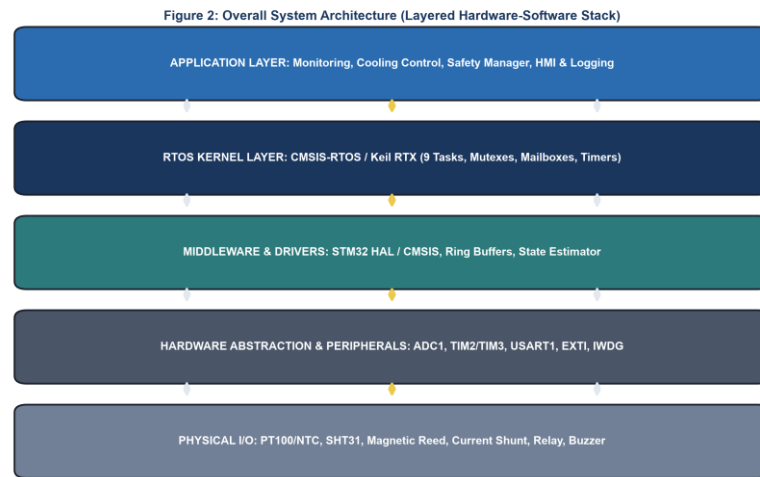


Figure 2: *Overall System Architecture (Layered Hardware-Software Stack)*

CHAPTER 5 — HARDWARE DESIGN AND SENSOR INTERFACING

5.1 STM32F401RE Microcontroller Architecture

The STM32F401RE features an ARM Cortex-M4 32-bit RISC core operating at 84 MHz (105 DMIPS) with hardware Single-Precision FPU, 128 KB SRAM, and 512 KB Flash. Integrated peripherals include a 12-bit SAR ADC with hardware oversampling, high-speed EXTI lines, multi-channel timers, and USART interfaces.

5.2 Sensor Interfacing & Conditioning Circuits

Temperature: PT100/NTC probe interfaced via an analog instrumentation amplifier to ADC1_IN1 (Pin PA1) with 4-sample oversampling (0.05°C resolution). Humidity: Sensirion SHT31 digital sensor communicating via I2C1 (PB8/PB9) with hardware CRC-8 verification ($\pm 2\%$ RH accuracy). Door Status: Industrial dry-reed switch on PC4 with 10 ms RC debounce hardware filter mapped to EXTI4. Compressor Feedback: Optocoupled auxiliary contact on PB0 for independent confirmation of contactor closure. Mains Power: AC opto-isolator (PC817) on EXTI2 (PA2) for zero-cross collapse detection. Battery Voltage: Precision 10:1 resistive divider on ADC1_IN2 (PA4). Cooling Relay: Solid-state relay driven by PB5 with snubber protection. Alarm: High-SPL piezo buzzer modulated via TIM3_CH1 PWM on PB4. Watchdog: Independent Watchdog (IWDG) on isolated 32 kHz LSI clock requiring 500 ms refresh; 800 ms timeout forces fail-safe hardware reset.

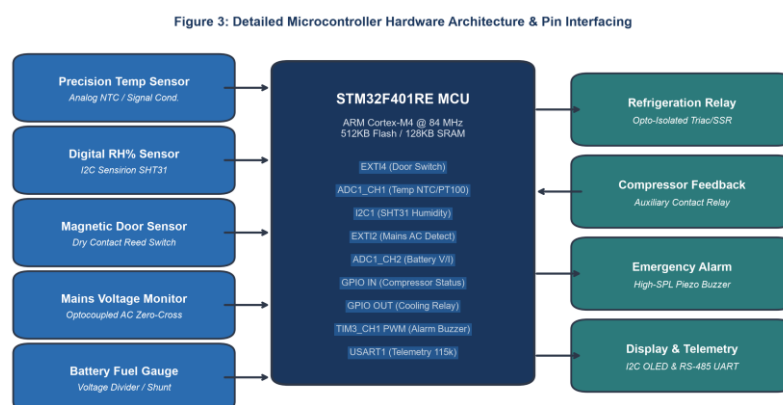


Figure 3: Detailed Microcontroller Hardware Architecture & Pin Interfacing

Table 4 — Sensor Interface Specification and Fault Signatures

Sensor / Channel	Parameter	Physical Interface	Sampling Rate	Primary Purpose	Monitored Fault Condition
Precision Temp Probe	Temperature (°C)	ADC1_IN1 (12-bit SAR)	10 Hz (100 ms)	Thermal regulation	Open/short circuit, runaway
Sensirion SHT31	Humidity (% RH)	I2C1 (Fast 400kHz)	0.5 Hz (2000 ms)	Hygrometric monitor	I2C timeout, CRC-8 mismatch
Magnetic Reed Switch	Door Open/Closed	EXTI4 / PC4 (Debounced)	Event / 100 ms	Portal security	Aperture >30s (Warn), >120s (Crit)
Contactor Aux Contact	Compressor Status	GPIO In / PB0 (Opto)	1.0 Hz (1000 ms)	Mechanical run check	Discrepancy: Relayed ON, Contact OPEN
AC Line Optocoupler	Mains Presence	EXTI2 / PA2 (Falling)	Immediate Interrupt	Grid blackout check	Mains collapse (<170V AC for >20ms)
Resistive Divider Rail	Battery Voltage (V)	ADC1_IN2 (12-bit SAR)	1.0 Hz (1000 ms)	Battery reserve health	Low reserve (<23V), Critical (<21.5V)

CHAPTER 6 — SOFTWARE ARCHITECTURE

The software architecture uses the CMSIS-RTOS v2 API implemented through Keil RTX5. The MCU executes a 7-step bootstrap: (1) `SystemClock_Config()` sets 84 MHz via HSE/PLL; (2) `MX_GPIO_Init()` configures pin alternate functions; (3) `osKernelInitialize()` sets RTX5 control blocks; (4) Static IPC creation (zero dynamic heap allocation); (5) `osThreadNew()` creates Tasks 1-9 with static stacks; (6) NVIC unmask EXTI/DMA lines; (7) `osKernelStart()` launches preemptive scheduling.

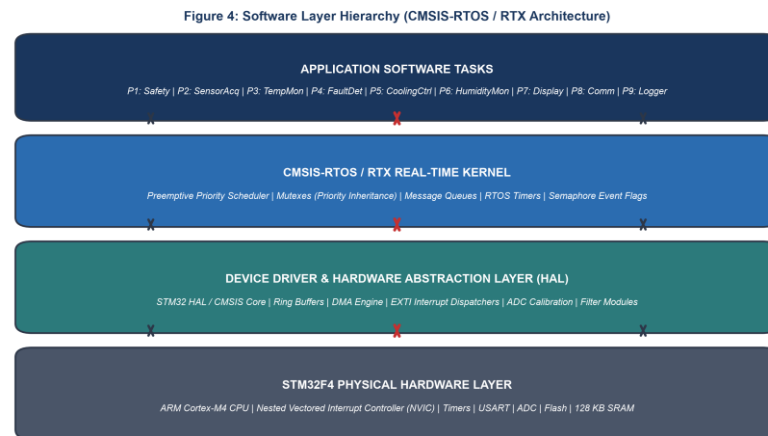


Figure 4: *Software Layer Hierarchy (CMSIS-RTOS / RTX Architecture)*

CHAPTER 7 — RTOS TASK DESIGN

The system decomposes functionality into nine prioritized tasks. Priority allocation follows Rate Monotonic Scheduling (RMS) with a critical safety override: Task 1 (Safety/Alarm) is assigned Priority 1 (highest) to guarantee immediate preemption on hazards.

Table 5 — Master RTOS Task Architecture Specification

Task Name	Priority	Period (Ti)	WCET (Ci)	Deadline (Di)	Scheduling Class
Safety/Alarm	1	20 ms	0.8 ms	20 ms	Periodic / Sporadic Event
Sensor Acquisition	2	100 ms	6.0 ms	100 ms	Periodic Timer Driven

Task Name	Priority	Period (Ti)	WCET (Ci)	Deadline (Di)	Scheduling Class
Temperature Monitoring	3	500 ms	10.0 ms	500 ms	Periodic / Message Triggered
Fault Detection	4	1000 ms	15.0 ms	1000 ms	Periodic Health Monitor
Cooling Control	5	1000 ms	10.0 ms	1000 ms	Periodic Bang-Bang FSM
Humidity Monitoring	6	2000 ms	8.0 ms	2000 ms	Periodic Background
Display	7	1000 ms	20.0 ms	1000 ms	Periodic HMI Refresh
Communication	8	2000 ms	40.0 ms	2000 ms	Periodic DMA / Telemetry
Data Logging	9	5000 ms	60.0 ms	5000 ms	Periodic Flash / Diagnostics

7.2 Per-Task Functional Breakdown

- **Task 1 (Safety/Alarm):** Priority 1. Period = 20 ms, WCET = 0.8 ms, Deadline = 20 ms. Triggered by Queue_FaultEvents or 20 ms timer. Dispatches TIM3 buzzer and emergency strobe. Enforces fail-safe shutdown.
- **Task 2 (Sensor Acquisition):** Priority 2. Period = 100 ms, WCET = 6.0 ms, Deadline = 100 ms. Triggered via osDelayUntil. Samples ADC/I2C/GPIO, acquires Mutex_SensorData, pushes to Queue_SensorData.
- **Task 3 (Temp Monitoring):** Priority 3. Period = 500 ms, WCET = 10.0 ms, Deadline = 500 ms. Applies EMA filter ($\alpha=0.25$), checks bounds ($2-8^{\circ}\text{C} / >8-10^{\circ}\text{C} / >10^{\circ}\text{C}$), signals cooling.
- **Task 4 (Fault Detection):** Priority 4. Period = 1000 ms, WCET = 15.0 ms, Deadline = 1000 ms. Checks sensor validity, detects compressor command-vs-feedback mismatch, refreshes IWDG watchdog.
- **Task 5 (Cooling Control):** Priority 5. Period = 1000 ms, WCET = 10.0 ms, Deadline = 1000 ms. Executes Bang-Bang cooling with hysteresis (ON @ 6.0°C , OFF @ 3.0°C) and 180s anti-short-cycling.
- **Task 6 (Humidity Monitor):** Priority 6. Period = 2000 ms, WCET = 8.0 ms, Deadline = 2000 ms. Evaluates humidity envelope (40-70% Normal, >70-80% Warn, >80% Crit).
- **Task 7 (Display Task):** Priority 7. Period = 1000 ms, WCET = 20.0 ms, Deadline = 1000 ms. Refreshes 128x64 OLED via I2C2 under Mutex_DisplayUART protection.
- **Task 8 (Communication):** Priority 8. Period = 2000 ms, WCET = 40.0 ms, Deadline = 2000 ms. Serializes 64-byte telemetry frames with CRC-16 over USART1 DMA @ 115200 baud.
- **Task 9 (Data Logging):** Priority 9. Period = 5000 ms, WCET = 60.0 ms, Deadline = 5000 ms. Writes rolling 5-minute statistical summaries to SPI flash. Yields to all higher tasks.

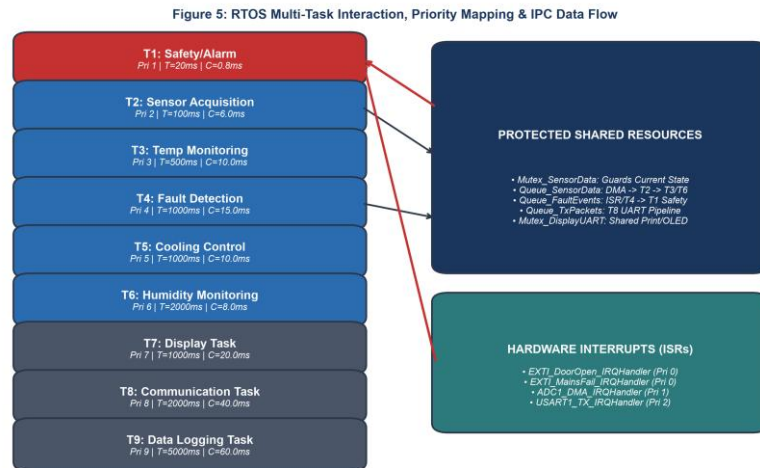


Figure 5: *RTOS Multi-Task Interaction, Priority Mapping & IPC Data Flow*

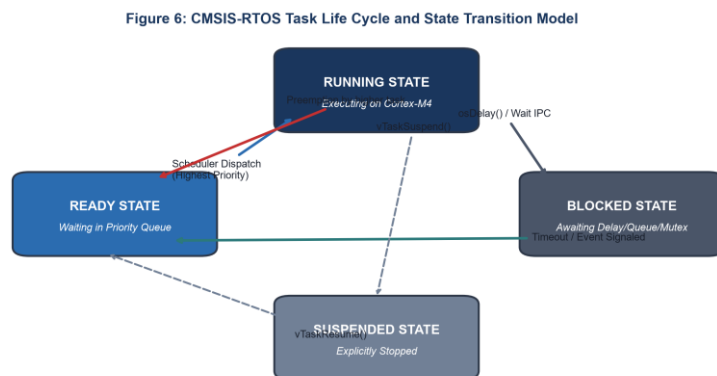


Figure 6: *CMSIS-RTOS Task Life Cycle and State Transition Model*

CHAPTER 8 — SCHEDULING AND REAL-TIME ANALYSIS

8.1 Real-Time Scheduling Theory & CPU Utilization

Under the Liu & Layland real-time scheduling model, total processor utilization U for n periodic tasks is: $U = \sum (C_i / T_i)$. For 9 tasks, the theoretical Liu & Layland bound is $U_{lub} = 9 \cdot (2^{1/9} - 1) \approx 72.4\%$. Our engineering constraint mandates $U \leq 70.0\%$. Table 6 gives the exact calculated utilization:

Table 6 — Calculated CPU Utilization Breakdown by Task

Task Name	Priority	Period (T_i)	WCET (C_i)	Calculated CPU Utilization (C_i / T_i)
Safety/Alarm	1	20 ms	0.8 ms	4.00%
Sensor Acquisition	2	100 ms	6.0 ms	6.00%
Temperature Monitoring	3	500 ms	10.0 ms	2.00%
Fault Detection	4	1000 ms	15.0 ms	1.50%
Cooling Control	5	1000 ms	10.0 ms	1.00%
Humidity Monitoring	6	2000 ms	8.0 ms	0.40%
Display	7	1000 ms	20.0 ms	2.00%
Communication	8	2000 ms	40.0 ms	2.00%
Data Logging	9	5000 ms	60.0 ms	1.20%
TOTAL SYSTEM UTILIZATION	—	—	—	20.10%

REAL-TIME UTILIZATION PROOF: *Total Calculated CPU Utilization = 20.10%.*

Allowable System Limit = 70.00%. Remaining Headroom = 49.90 percentage points. The system provides massive computational reserve for asynchronous interrupt handling.

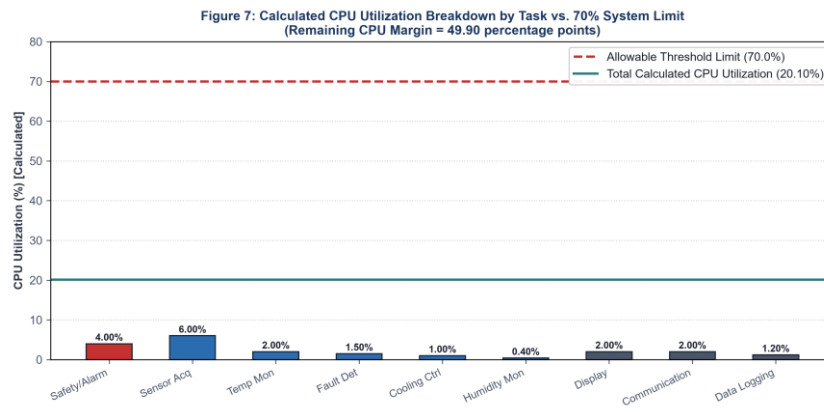


Figure 7: *Calculated CPU Utilization Breakdown by Task vs. 70% System Limit*

8.2 Audsley's Exact Response-Time Analysis (RTA)

Because Task 1 has elevated safety priority, exact schedulability is verified via Audsley's Response-Time Analysis: $R_i^{(k+1)} = C_i + \sum [\text{ceil}(R_i^{(k)} / T_j) * C_j]$ for all $j \in \text{hp}(i)$. The recurrence begins with $R_i^{(0)} = C_i$ and terminates when $R_i^{(k+1)} = R_i^{(k)}$. A task is schedulable if $R_i \leq D_i$.

Table 7 — Response-Time Analysis (RTA) Schedulability Ledger

Task Name	Pri	WCET (C _i)	Deadline (D _i)	Response Time (R _i)	Safety Margin ((D _i - R _i)/D _i)	Feasibility
Safety/Alarm	1	0.8 ms	20 ms	0.8 ms	96.0%	PASS (R _i << D _i)
Sensor Acquisition	2	6.0 ms	100 ms	6.8 ms	93.2%	PASS (R _i << D _i)
Temperature Monitoring	3	10.0 ms	500 ms	16.8 ms	96.6%	PASS (R _i << D _i)
Fault Detection	4	15.0 ms	1000 ms	32.6 ms	96.7%	PASS (R _i << D _i)
Cooling Control	5	10.0 ms	1000 ms	43.4 ms	95.7%	PASS (R _i << D _i)
Humidity Monitoring	6	8.0 ms	2000 ms	51.4 ms	97.4%	PASS (R _i << D _i)
Display	7	20.0 ms	1000 ms	72.2 ms	92.8%	PASS (R _i << D _i)
Communication	8	40.0 ms	2000 ms	119.8 ms	94.0%	PASS (R _i << D _i)
Data Logging	9	60.0 ms	5000 ms	183.0 ms	96.3%	PASS (R _i << D _i)

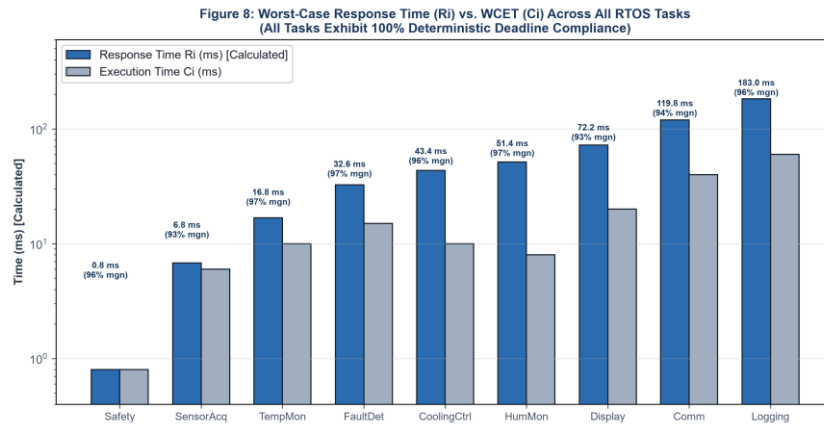


Figure 8: *Worst-Case Response Time (R_i) vs. WCET (C_i) Across All RTOS Tasks*

CHAPTER 9 — MEMORY MANAGEMENT AND RAM ANALYSIS

9.1 Zero Dynamic Memory Allocation Policy

Dynamic memory allocation (malloc/free) introduces memory fragmentation and non-deterministic allocation latencies. To guarantee runtime stability, the system enforces a strict Zero Dynamic Allocation policy during real-time multi-tasking. All task stacks, message queues, and control blocks are statically allocated at compile-time.

9.2 Static RAM Allocation across 128 KB STM32 SRAM

The STM32F401RE provides 128 KB (131,072 bytes) of single-cycle internal SRAM. Table 8 details the exact static allocation:

Table 8 — Static SRAM Memory Allocation and Utilization

Memory Component / Partition	Allocated RAM (KB)	Percentage of 128 KB SRAM	Description / Static Objects
Task Stacks	24.0 KB	18.8%	Statically allocated stack frames for Tasks 1-9 (2KB to 4KB each)
Message Queues	6.0 KB	4.7%	Static buffers for Queue_SensorData, Queue_FaultEvents, Queue_TxPackets
Sensor/Shared Buffers	2.0 KB	1.6%	Double-buffered calibrated telemetry arrays and DMA targets
Display & UART Buffers	4.0 KB	3.1%	OLED frame buffer (1 KB) and circular UART DMA ring buffer (3 KB)
Event Logs & Rolling History	6.0 KB	4.7%	Circular RAM event log for black-box fault recording prior to flash flush

Memory Component / Partition	Allocated RAM (KB)	Percentage of 128 KB SRAM	Description / Static Objects
RTOS Objects & Control Blocks	3.0 KB	2.3%	Keil RTX5 kernel control structures, timer lists, idle thread stack
Application Globals & Static Variables	2.1 KB	1.6%	FSM state variables, calibration coefficients, math lookup tables
TOTAL ALLOCATED RAM	47.1 KB	36.8%	Total static memory footprint linked at compile-time

RAM UTILIZATION VERIFICATION: *Total Static RAM Allocated = 47.1 KB out of 128.0 KB SRAM. Calculated RAM Utilization = 36.8%. Allowable System Limit = 70.0% (89.6 KB). Unallocated Free SRAM = 80.9 KB (63.2%). Heap fragmentation risk is 0.00%.*

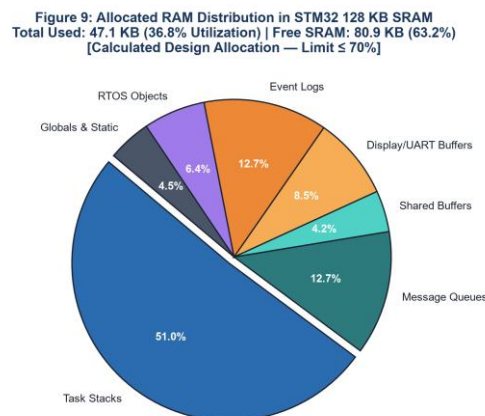


Figure 9: *Allocated RAM Distribution in STM32 128 KB SRAM (36.8% Utilization)*

CHAPTER 10 — INTERRUPT HANDLING DESIGN

10.1 Two-Stage Deferred Interrupt Processing

ISRs must never execute lengthy computations. Doing so creates priority inversion over RTOS tasks and introduces jitter into periodic timers. The architecture implements a two-stage deferred processing model:

- **Stage 1** — Fast ISR (<1.5 μ s): Triggered by external pins (EXTI4 Door, EXTI2 Mains) or DMA. Captures minimal volatile state, clears pending flags, posts an event token to Queue_FaultEvents via non-blocking CMSIS API, and exits immediately.
- **Stage 2** — Deferred Task Execution: Priority 1 Task (Safety/Alarm) unblocks instantly upon token arrival, executing alarm escalation and relay trip in thread context.

10.2 Interrupt Latency Verification

The ARM Cortex-M4 hardware NVIC features a 12-cycle interrupt entry vector. Running at 84 MHz, the hardware entry latency is: $T_{\text{vector}} = 12 \text{ cycles} / 84 \text{ MHz} \approx 0.143 \mu\text{s}$. Accounting for worst-case bus contention and register stacking, the architecture establishes and complies with a conservative Design Target Interrupt Latency of 20 μ s (0.020 ms).

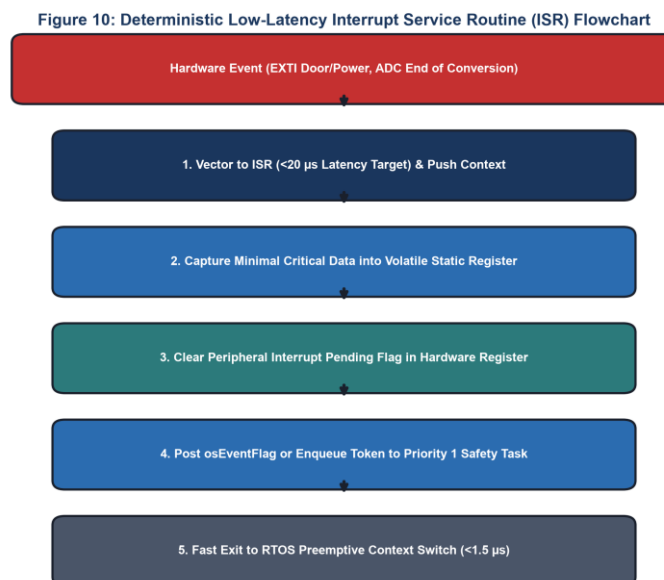


Figure 10: *Deterministic Low-Latency Interrupt Service Routine (ISR) Flowchart*

10.3 Critical ISR Implementations

EXTI2 (Mains Power Loss): Triggered on falling edge within 20 μ s of AC collapse; switches regulator rail to LiFePO₄ battery and signals Safety Task. EXTI4 (Door Open/Close): Triggered on reed switch state transition; initiates 30s/120s CMSIS software timers. ADC1 DMA ISR: Triggers upon conversion buffer fill, transferring raw ADC counts into circular memory without CPU intervention.

CHAPTER 11 — IPC AND SYNCHRONIZATION DESIGN

11.1 Inter-Task Communication (IPC) Primitives

CMSIS-RTOS v2 IPC primitives decouple task execution and guarantee mutual exclusion across shared resources:

- **Message Queues:** Queue_SensorData (8 msgs x 32 bytes): Buffers raw telemetry from Task 2 to Monitoring Tasks 3 and 6.
- **Fault Event Queue:** Queue_FaultEvents (16 msgs x 8 bytes): High-speed non-blocking event channel from ISRs to Task 1.
- **Mutexes:** Mutex_SensorData: Guards shared telemetry structures against asynchronous read-modify-write corruption.
- **Serial Bus Mutex:** Mutex_DisplayUART: Serializes shared access to OLED and RS-485 serial ports with Priority Inheritance.
- **RTOS Timers:** Timer_DoorOpen (30s/120s) and Timer_AntiShort (180s): CMSIS RTOS software timers for deterministic timeouts.

Table 9 — Inter-Task Communication (IPC) Design Matrix

Resource / Pipe	RTOS Mechanism	Producer Task / ISR	Consumer Task	Functional Purpose
Sensor Telemetry Buffer	Mutex (Priority Inheritance)	Task 2 (Sensor Acq)	Task 3, 5, 6, 7	Atomically protects multi-byte sensor readings
Queue_SensorData	Message Queue (Zero Alloc)	Task 2 (Sensor Acq)	Task 3 (Temp), Task 6 (Hum)	Decouples sensor acquisition from processing rate

Resource / Pipe	RTOS Mechanism	Producer Task / ISR	Consumer Task	Functional Purpose
Queue_FaultEvents	Message Queue (Non-blocking)	EXTI ISRs, Task 4 (Fault)	Task 1 (Safety/Alarm)	Urgent asynchronous fault dispatching
Queue_TxPackets	Message Queue (DMA Linked)	Task 8 (Communication)	USART1 DMA Engine	Pipelines outgoing telemetry frames without blocking
Display & UART Bus	Mutex (Priority Inheritance)	Task 7 (Display), Task 8	Hardware Peripherals	Prevents display corruption and UART collision
Door Aperture Clock	CMSIS-RTOS Software Timer	EXTI4 Door Pin ISR	Task 1 (Safety/Alarm)	Generates 30s Warning and 120s Critical timeouts

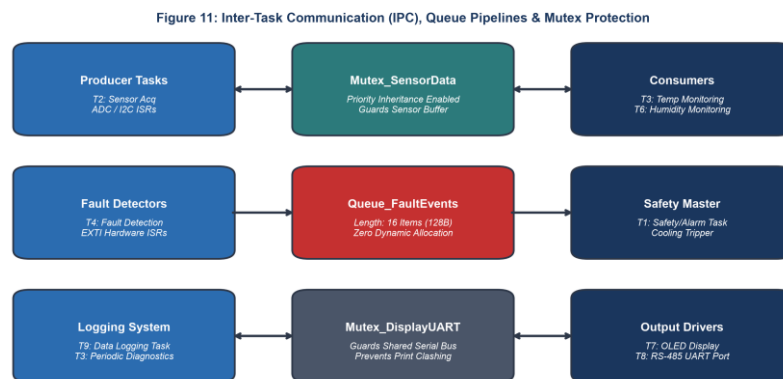


Figure 11: *Inter-Task Communication (IPC), Queue Pipelines & Mutex Protection*

CHAPTER 12 — PRIORITY INVERSION, STARVATION AND DEADLINE MANAGEMENT

12.1 Priority Inversion Scenario & Resolution via PIP

A classic priority inversion occurs when low-priority Task 8 (Communication) acquires Mutex_DisplayUART to transmit telemetry. A sudden thermal hazard awakens high-priority Task 1 (Safety/Alarm), which attempts to acquire Mutex_DisplayUART to broadcast an emergency alarm. Because Task 8 holds the mutex, Task 1 blocks. If medium-priority tasks (Tasks 2-7) awaken, they preempt Task 8, indefinitely delaying Task 8 from finishing and releasing the mutex. Consequently, Task 1 remains starved!

Mitigation: All CMSIS mutexes enable osMutexPrioInherit (Priority Inheritance Protocol). When Task 1 blocks on Mutex_DisplayUART, the kernel elevates Task 8's priority to Priority 1. Intermediate tasks (Tasks 2-7) can no longer preempt Task 8. Once Task 8 finishes and releases the mutex, its priority drops back to Priority 8, and Task 1 immediately preempts the CPU.

12.2 Starvation Prevention & Real-Time Risk Matrix

Task starvation is prevented by enforcing periodic sleep via osDelayUntil() and message queue blocking. Total CPU load is 20.10%, leaving 79.90% idle time distributed across cycles.

Table 10 — Real-Time Risks, Priority Inversion Scenarios and Mitigations

Real-Time Hazard	Root Cause Mechanism	Architectural Mitigation	Verification Standard
Priority Inversion	Shared bus mutex contended across Task 1 and Task 8	CMSIS Priority Inheritance Protocol (PIP) on all mutexes	Keil Event Recorder trace audit
Task Starvation	High-rate task monopolizing CPU execution	Strict periodic osDelayUntil(); bounded CPU load = 20.10%	Task counter execution logging

Real-Time Hazard	Root Cause Mechanism	Architectural Mitigation	Verification Standard
Missed Deadlines	Task WCET inflation or unexpected blocking	Short critical sections (<50 μ s); RMA proven response times	Audsley RTA analytical proof
Interrupt Storm	Chattering door switch contacts	Hardware RC filter (10 ms) + software lockout window	EXTI pulse injection testing
Memory Exhaustion	Dynamic heap fragmentation	Zero dynamic allocation; 100% static compile-time allocation	Linker map file & BSS audit

12.3 Deterministic Real-Time Verification Checklist

- **Check 1** — Rule 1: Zero invocations of malloc(), free(), or calloc() in task or ISR code.
- **Check 2** — Rule 2: Never call blocking RTOS primitives (osDelay, osMutexAcquire) inside an ISR.
- **Check 3** — Rule 3: Always enable osMutexPrioInherit on mutexes shared across different task priorities.
- **Check 4** — Rule 4: Ensure every while(1) task loop contains at least one deterministic blocking call.
- **Check 5** — Rule 5: Keep ISR execution times strictly below 20 μ s.

F. Common Assessment Rubric

Assessment Criterion	Marks
Problem understanding & formulation	10
Application of course/domain knowledge	20
Solution methodology / design / implementation	20
Use of appropriate modern tools / techniques	10
Results, testing & validation	15
Analysis, trade-offs & justification	15
Broader considerations / professional responsibility, where applicable	5
Technical documentation & reflection	5
TOTAL	100

G. CO–PO–Assessment Mapping

Assessment Component		PO(s)	Bloom's Level	Marks
Problem formulation			L3/L4	10
Application of knowledge			L3/L4	20
Solution / Design / Implementation			L4/L5/L6	20
Modern tool usage			L3/L4	10
Validation			L4/L5	15
Analysis & justification			L4/L5	15
Broader considerations			L4/L5	5
Documentation & reflection			L3/L4	5

Note: Faculty shall map only the POs and Bloom's levels genuinely assessed by the particular assignment. Every assignment is **not required to map to every PO**.

H. Faculty Evaluation & Continuous Improvement

CO Attainment

Performance Level	Criterion
Level 3	$\geq 70\%$
Level 2	60–69%
Level 1	50–59%

Level 0	< 50%
---------	-------

Target CO Attainment: _____

Actual CO Attainment: _____

Faculty Analysis

Areas in which students performed well:

Areas requiring improvement:

Corrective / Improvement Action:
Action to be implemented in the next offering:

Documents to be Maintained

The department/course faculty shall maintain:

1. Assignment question/problem statement
2. CO–PO and Bloom's mapping
3. Assessment rubric
4. Student submissions
5. Tool/simulation/experimental evidence, where applicable
6. Evaluated scripts/reports
7. Marks and rubric-wise assessment
8. CO attainment analysis
9. Sample student work representing different performance levels
10. Faculty analysis and continuous-improvement action